How Much Does the Spatial Module Really Help? A Systematic Analysis for Long-Term MTS Forecasting

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Abstract. Recent studies shown that simple linear models can match or outperform both spatio-temporal graph neural network (STGNN) and Transformer approaches for long-term multivariate time series forecasting (LTSF) on diverse benchmark datasets. While STGNNs commonly use Time-Then-Space (TTS) architectures to capture inter-variable dependencies, these architectures have historically relied on complex temporal components such as Temporal Convolutional Networks (TCNs) and Recurrent Neural Networks (RNNs) to learn temporal dynamics. This study systematically analyzes the effect of adding spatial modules to three linear backbones (DLinear, NLinear, and RLinear) and PatchTST across four benchmark datasets. The spatial modules are evaluated together with standard architectural components used in TTS forecasting models, including prediction heads and residual skip connections. Our experiments reveal that the effect of spatial augmentation depends strongly on the backbone and dataset, and that the best-performing configuration varies across models and datasets. These findings provide practical guidance for the design and evaluation of spatial modules in LTSF. Source code is available at https://github.com/htmoges/spatial-augmentation-ltsf.

Keywords: Long-term time series forecasting · Spatial Module · Spatio-temporal models · Time-Then-Space architectures

1 Introduction

Spatio-temporal graph neural networks (STGNNs) combine spatial graph convolutions with temporal modules and have been widely applied to multivariate time series (MTS) forecasting [22,12,21,1,20,8]. A standard architectural layout in these models is the Time-Then-Space (TTS) design. In this approach, a temporal module is first used to capture temporal dynamics, which are subsequently fed into a spatial module to capture dependencies between variables [6,4].

These architectures historically relied on complex temporal components such as Temporal Convolutional Networks (TCNs) [11] or Recurrent Neural Networks (RNNs) [7] to learn these temporal dynamics before spatial processing.

The long-term MTS forecasting (LTSF) landscape has since shifted towards much simpler architectures. Studies have demonstrated that simple linear models, such as DLinear, NLinear, and RLinear, can match or substantially outperform more complex STGNNs and Transformer models on standard LTSF benchmarks [23,13]. At the same time, lightweight graph based spatial approaches have begun to appear in the LTSF domain [2,17]. In this setting, spatial modules are typically evaluated by comparing the spatio-temporal architecture against the temporal backbone alone.

However, spatial augmentation is commonly introduced together with additional architectural components such as prediction heads, normalization, gating, and residual skip connections. Consequently, improvements observed in the full architecture cannot be directly associated with the spatial module itself. This makes it difficult to determine whether the addition of a spatial module consistently improves forecasting performance across different backbones and datasets.

To address this problem, we systematically explore the addition of a spatial module to three dominant linear backbones, DLinear, NLinear, and RLinear, together with the strong Transformer baseline PatchTST. We study four architectural configurations: **L** (backbone only), **L+PH** (prediction head added), **L+PH+R** (prediction head and input skip connection), and **L+PH+R+S** (full TTS architecture). The **L+PH+R** configuration serves as the reference configuration for evaluating the spatial module, since it preserves the additional fusion components while removing only the spatial module itself. For PatchTST, we additionally evaluate the complete $\{$Spatial, No-Spatial$\}\times\{$Residual, No-Residual$\}$ configurations to analyze the role of the residual connection when applying a spatial module.

Our findings are specific to one type of spatial augmentation: learned-adjacency spatial modules of the form $(\mathbf{A} - \mathbf{I})\mathbf{Z}$ with gated residual fusion [17]. More broadly, however, the need to compare multiple architectural configurations applies whenever a spatial module is introduced with additional non-spatial components.

Our key contributions are:

1. **Exploration of spatial augmentation for linear LTSF models.** Across three linear backbones and PatchTST on four datasets, our analysis reveals that while the STGNN architecture often outperforms the baseline, the specific effect of the spatial module varies substantially, with performance varying across different backbones and datasets.
2. **Evaluation of TTS architectural components.** By comparing multiple architectural configurations, we show that prediction heads and residual skip connections often account for much of the observed forecasting improvement. Furthermore, a controlled study using PatchTST shows that applying a spatial module without a residual skip connection reduces forecasting performance in the tested settings.

2 Related Work

Recent work has shown that comparatively simple architectures can be highly competitive for LTSF forecasting. Zeng et al. [23] showed that linear models such as DLinear and NLinear can match or outperform several Transformer-based models on standard LTSF benchmarks. RLinear [13] further improves linear forecasting through reversible instance normalization [9], while PatchTST [18] achieves strong performance by combining channel-independent modeling with patched Transformer representations. Furthermore, architectures such as SegRNN [14] and TSMixer [5] demonstrate that efficient temporal segmentation and lightweight MLP-based post-processing can also achieve highly competitive results. These findings have shifted attention toward lightweight temporal backbones, raising the question of whether additional spatial modules still provide consistent benefit when attached to already strong LTSF models.

Spatiotemporal graph neural networks (STGNNs) commonly follow a Time-Then-Space (TTS) architecture, where temporal representations learned by modules such as TCNs or RNNs are subsequently processed by a spatial module to capture dependencies between variables [8,6]. Representative architectures including STGCN, DCRNN, Graph WaveNet, AGCRN, MTGNN, and StemGNN [22,12,21,1,20,3] have been especially successful in short-term traffic forecasting, where sensor networks often provide a meaningful physical graph. Models such as GTS [19] jointly learn graph structure and forecasting, while SimST [15] replaces full graph message passing with simplified local and global spatial learning modules while retaining competitive performance. These developments established TTS architectures as a standard framework for spatiotemporal forecasting and motivated the introduction of similar spatial extensions into the LTSF domain.

Graph-based spatial modules have now begun to appear in the LTSF domain. Unlike traffic forecasting benchmarks, standard LTSF datasets typically do not provide an externally specified sensor or physical graph, requiring spatial modules to learn inter-variable structure directly from data. Approaches such as ForecastGrapher [2] and Lite-STGNN [17] introduce graph based spatial structure into LTSF architectures following the broader TTS design used in STGNNs.

However, these spatial modules are typically introduced together with additional architectural components such as prediction heads, normalization, gating, or residual skip connections. As a result, it remains unclear whether observed forecasting improvements arise from the spatial module itself or from the accompanying components. This limitation is particularly critical now that strong linear backbones already perform competitively on standard LTSF benchmarks. We address this gap by systematically evaluating spatial modules across multiple architectural configurations and forecasting backbones.

3 Methodology

3.1 Spatial Augmentation Framework

Our framework adds a spatial module on top of a pre-trained temporal backbone, following the lightweight TTS design used in recent LTSF architectures [17]. Motivated by recent evidence that lightweight MLP-based post-processing can itself contribute substantially to forecasting performance [5,18], we also include a prediction head between the backbone and the spatial module. The prediction head provides a lightweight nonlinear transformation of the backbone outputs and serves as an additional architectural component within the evaluation framework.

Standard backbone versus full model comparisons cannot separate the effects of the spatial module from those of accompanying architectural components. We therefore evaluate multiple architectural configurations that introduce prediction heads, residual skip connections, and spatial modules within a shared forecasting pipeline.

Given multivariate input $\mathbf{X} \in \mathbb{R}^{N \times L}$ (N channels, L time steps) and backbone predictions $\hat{\mathbf{Y}}_{\text{base}} \in \mathbb{R}^{N \times H}$ (H forecast horizon), the full pipeline is:

1. **Prediction head**
$$\mathbf{Z} = \text{MLP}(\hat{\mathbf{Y}}_{\text{base}}),$$
where the MLP is a two-layer network with GELU activation applied independently to each channel.
2. **Spatial message passing**
$$\mathbf{S} = \sigma(\mathbf{g}) \odot [(\mathbf{A} - \mathbf{I}) \cdot \text{LN}(\mathbf{Z})],$$
where $\mathbf{A}$ is a learned adjacency matrix, LN denotes layer normalization, and $\sigma(\mathbf{g})$ is a learned gate initialized near zero.
3. **Residual fusion**
$$\hat{\mathbf{Y}} = \mathbf{Z} + \mathbf{S} + \hat{\mathbf{Y}}_{\text{base}},$$
where the final term is the input skip connection.

For PatchTST, the prediction head is omitted because the Transformer encoder already contains nonlinear layers. Accordingly, only the spatial module and residual configurations are varied for PatchTST.

Figure 1(a) illustrates the full TTS architecture, where the prediction head and spatial module operate on top of the backbone forecast before residual fusion with the input skip connection. Figure 1(b) summarizes the evaluation configurations used throughout the analysis. By selectively activating or removing the prediction head (PH), residual connection (R), and spatial module (S), the effects of individual components and their interactions can be analyzed within a shared architecture.

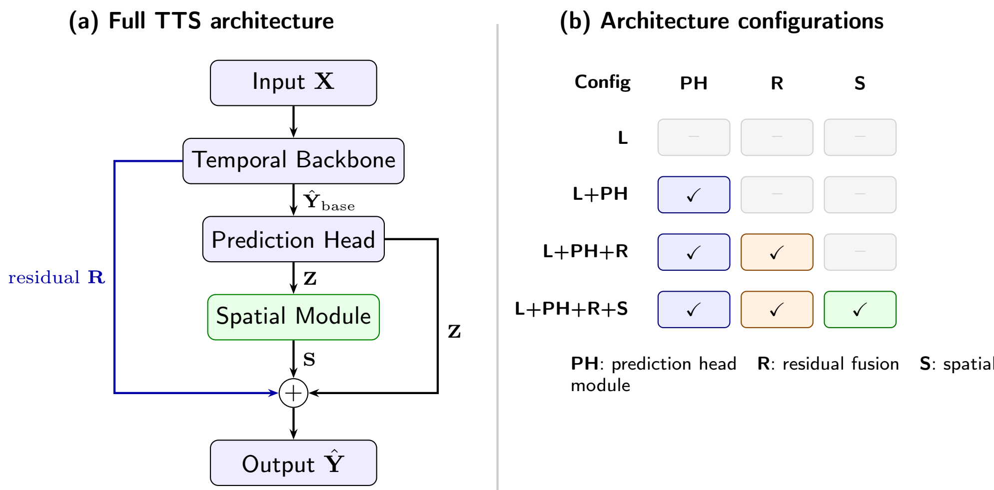

Fig. 1: TTS architecture and evaluation configurations. (a) Full spatio-temporal pipeline. (b) Component activation patterns used to evaluate the distinct effects of the prediction head, residual connection, and spatial module.

3.2 Evaluation Configurations

To analyze the effects of different architectural components, we evaluate multiple configurations within the common architecture shown in Figure 1. We use the following shorthand throughout the paper.

Table 1: Architecture configurations used in the evaluation framework.

Config	Description	Role
L	Backbone only	Baseline
L+PH	Prediction head added	Prediction head
L+PH+R	Prediction head + residual	Residual fusion
L+PH+R+S	Full TTS architecture	Spatial augmentation

All relative changes are normalized by the baseline MSE(L):

$$\text{Res\%} = \frac{\text{MSE(L+PH+R)} - \text{MSE(L+PH)}}{\text{MSE(L)}} \times 100 \tag{1}$$

$$\text{Sp\%} = \frac{\text{MSE(L+PH+R+S)} - \text{MSE(L+PH+R)}}{\text{MSE(L)}} \times 100 \tag{2}$$

$$\text{Gain\%} = \frac{\text{MSE(L+PH+R+S)} - \text{MSE(L)}}{\text{MSE(L)}} \times 100 \tag{3}$$

Because all three quantities share the same denominator, the decomposition is additive:

$$\mathrm{PH\%} + \mathrm{Res\%} + \mathrm{Sp\%} = \mathrm{Gain\%}.$$

The L+PH+R configuration therefore serves as the reference configuration for evaluating the effect of the spatial module while preserving the remaining non-spatial components.

PatchTST configuration study. For PatchTST, we additionally evaluate the complete {Spatial, No-Spatial} × {Residual, No-Residual} crossing shown in Table 2. Because PatchTST already contains nonlinear transformations, there is no separate prediction head to vary. The Base configuration therefore corresponds to L in the linear protocol.

Table 2: PatchTST architectural configurations.

Config	Spatial	Residual	Description
Base	×	×	Pure PatchTST baseline
+R	×	✓	Residual only
+R+S	✓	✓	Spatial + residual
+S	✓	×	Spatial only (stress test)

3.3 Spatial Module

The adjacency matrix is parameterized following [20] as

$$\mathbf{A} = \mathrm{TopK}\big(\mathrm{softmax}(\mathbf{E}_1\mathbf{E}_2^\top)\big),$$

where $\mathbf{E}_1, \mathbf{E}_2 \in \mathbb{R}^{N\times r}$ are low-rank embeddings with $r \ll N$. TopK retains the k largest entries per row, yielding a sparse and asymmetric adjacency. Because standard LTSF benchmarks typically do not provide predefined graphs, the adjacency is learned directly from data.

3.4 Backbone Architectures

We evaluate three linear backbones together with PatchTST as a strong Transformer baseline. The selected models differ in normalization strategy and model capacity.

The linear backbones include DLinear [23] (seasonal trend decomposition with shared linear projection), NLinear [23] (last value normalization with linear projection), and RLinear [13] (RevIN normalization [9] with linear projection). We additionally evaluate PatchTST [18], a channel-independent patched Transformer architecture with RevIN normalization.

4 Experimental Setup

4.1 Datasets and Dataset Profiling

Following a comparative empirical analysis perspective [10], we characterize the benchmark datasets and relate component behavior to measurable dataset properties. We evaluate on four standard LTSF benchmarks summarized in Table 3. Traffic exhibits the highest mean pairwise correlation ($|\rho| = 0.59$), suggesting the greatest potential for cross-channel interaction. ETTh1 (Electricity Transformer Temperature Hourly) shows the weakest overall correlation ($|\rho| = 0.24$) across only seven semantically heterogeneous variables, limiting the amount of coherent cross-channel structure available for spatial augmentation. Weather exhibits a broad correlation distribution (IQR $0.04 - -0.47$), reflecting a mixture of tightly coupled and weakly related channels.

The datasets also differ substantially in stationarity. Traffic is fully stationary under the augmented Dickey–Fuller test, while ETTh1 contains the lowest proportion of stationary channels.

Table 3: Dataset characteristics and profiling statistics. $|\rho|$ denotes the mean pairwise absolute Pearson correlation (estimated using a 50-channel subsample for Electricity and Traffic). ADF% reports the percentage of stationary channels under the augmented Dickey–Fuller test ($p < 0.05$).

Dataset	N	T	Freq	Mean $\|\rho\|$	Median $\|\rho\|$	IQR $\|\rho\|$	ADF%
Electricity	321	26,304	1h	0.49	0.48	0.24–0.77	66%
Traffic	862	17,544	1h	0.59	0.62	0.48–0.72	100%
Weather	21	52,696	10min	0.29	0.15	0.04–0.47	90%
ETTh1	7	17,420	1h	0.24	0.16	0.07–0.33	57%

4.2 Training and Evaluation Protocol

All experiments follow the standard LTSF evaluation protocol [18,23] using chronological 70/10/20 train, validation, and test splits, with MSE as the optimization objective. All models use input length $L = 96$ and forecast horizon $H = 720$. We additionally evaluate horizons $H \in \{96, 192, 336, 720\}$. For PatchTST, we also evaluate $L = 336$ to examine whether the spatial module behaves differently with longer input context.

Training uses Adam optimization with a learning rate 8×10^{-4}, early stopping with patience 10, and batch size 32 (16 for Traffic due to memory constraints). Each backbone retains its original normalization strategy (Section 3), while all variables are standardized using statistics computed from the training split.

Key comparisons are repeated across four independent random seeds using paired configurations with identical splits and initialization per seed. We report mean ± standard deviation, paired two-tailed t-tests across seeds, and paired effect sizes computed as $d_z = \bar{d}/s_d$. Relative changes are reported as percentages normalized by the backbone baseline. Negative percentages therefore indicate improved forecasting performance.

5 Results

In this section, we present the empirical results of the architectural evaluation. We first evaluate component-wise forecasting performance across three linear backbones (Section 5.1), and then analyze the behavior of the spatial module through a configuration study using PatchTST (Section 5.2). Finally, we examine how residual and spatial effects vary across forecast horizons and input lengths (Section 5.3).

5.1 Component-wise Analysis Across Linear Backbones

Table 4 summarizes forecasting performance across the three linear backbones and four datasets for the longest horizon setting ($H = 720$). Integrating a spatial module improves over the backbone baseline in 11 of the 12 backbone and dataset combinations, with DLinear on ETTh1 as the only exception. Every backbone benefits on at least three of the four datasets, although the strongest configuration varies substantially across models and datasets.

The best-performing configuration depends strongly on the backbone. NLinear most often benefits from the full TTS configuration (L+PH+R+S), whereas RLinear frequently performs best with the residual configuration (L+PH+R). DLinear shows more mixed behavior, particularly on Weather and ETTh1, where the prediction head alone already achieves the strongest result. Notably, the overall best forecasting performance across all four datasets († in Table 4) is consistently achieved by extended RLinear configurations (L+PH or L+PH+R), rather than by the full spatial configuration. Across all four datasets, the evaluated configurations substantially improve over the corresponding backbone baselines and consistently outperform the strongest standard linear models.

Figure 2 decomposes the total improvement into prediction head, residual, and spatial contributions. Panel (a) shows that the prediction head alone consistently improves forecasting performance, with gains ranging from approximately -0.6% to -13.5% relative to the backbone baseline. Panels (b) and (c), however, reveal strong backbone dependence in the residual and spatial effects.

The effect of the spatial module is beneficial for NLinear across all datasets, mixed for DLinear, and detrimental for RLinear on every dataset. Residual effects also vary substantially across datasets. On Electricity, the residual improves performance for all three backbones, whereas on Traffic it helps DLinear and RLinear but strongly degrades NLinear. In the NLinear Electricity setting, where both residual and spatial effects are beneficial, the residual connection accounts for most of the improvement between L+PH and the full TTS architecture. The beneficial spatial effects for NLinear on Traffic and Weather are also statistically significant across seeds ($p < 0.01$).

Table 4: Forecasting performance across linear backbones and datasets (MSE@720, mean ± std over four seeds). Bold indicates the best configuration for each backbone and dataset pair; underlined indicates second best. † marks the best overall linear configuration for each dataset.

Model	Dataset	L	L+PH	L+PH+R	L+PH+R+S
DLinear	Electricity	0.2433±0.0004	0.2350±0.0009	<u>0.2299±0.0003</u>	**0.2287±0.0028**
	Traffic	0.6473±0.0013	0.5950±0.0037	**0.5819±0.0009**	<u>0.5923±0.0032</u>
	Weather	0.3500±0.0018	**0.3340±0.0014**	0.3387±0.0051	<u>0.3368±0.0024</u>
	ETTh1	<u>0.6469±0.0031</u>	**0.6428±0.0068**	0.6500±0.0048	0.6536±0.0030
NLinear	Electricity	0.2545±0.0005	0.2366±0.0024	<u>0.2311±0.0009</u>	**0.2288±0.0016**
	Traffic	0.6474±0.0004	**0.5807±0.0066**	0.6284±0.0015	<u>0.5947±0.0043</u>
	Weather	0.3657±0.0006	<u>0.3416±0.0041</u>	0.3498±0.0013	**0.3347±0.0011**
	ETTh1	0.7054±0.0009	<u>0.6451±0.0086</u>	0.6457±0.0049	**0.6416±0.0066**
RLinear	Electricity	0.2541±0.0001	0.2348±0.0004	**0.2269±0.0003**†	<u>0.2317±0.0044</u>
	Traffic	0.6445±0.0009	<u>0.5574±0.0007</u>	**0.5482±0.0018**†	0.5925±0.0050
	Weather	0.3636±0.0003	**0.3313±0.0013**†	<u>0.3337±0.0057</u>	0.3367±0.0051
	ETTh1	0.7017±0.0012	0.6461±0.0126	**0.6351±0.0114**†	<u>0.6375±0.0069</u>

These trends are consistent with the dataset profiling statistics in Table 3. ETTh1, which has the weakest cross-channel correlation ($|\rho| = 0.24$), shows negligible or harmful spatial effects across all backbones. By contrast, Traffic and Electricity exhibit greater potential for spatial benefit, although the direction and magnitude of these effects still depend substantially on the temporal backbone and its normalization strategy.

Overall, the results show that spatial augmentation can substantially improve linear forecasting backbones, but the strongest configuration depends critically on how the spatial module interacts with the residual connection and the normalization strategy used by the temporal backbone.

5.2 PatchTST Configuration Analysis

Table 5 presents the complete PatchTST 2×2 interaction study across four datasets. The spatial-only configuration (+S), which applies the spatial module without the residual connection, degrades forecasting performance on every dataset. The degradation is especially large on Electricity (+15.0%) and ETTh1 (+15.6%), with consistently harmful effects also observed on Traffic (+6.8%) and Weather (+7.1%). These results indicate that spatial augmentation alone is not sufficient for stable forecasting within this architectural design.

By contrast, the anchored effect of the spatial module, measured by comparing +R and +R+S, is relatively small and dataset dependent. The spatial module modestly improves Traffic and Weather, while remaining nearly neutral on Electricity and ETTh1. The beneficial spatial effects on Traffic are also statistically significant across seeds ($p < 0.01$). This contrast between +S and +R+S suggests that the spatial module behaves very differently when applied independently versus when applied with a residual correction.

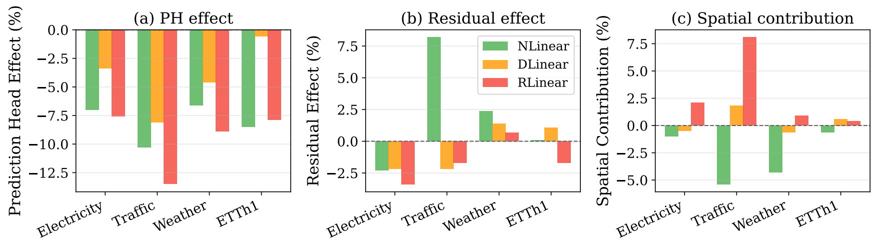

Fig. 2: Component-wise analysis across models and datasets (MSE@720). (a) Prediction head effect relative to L. (b) Residual effect relative to L+PH. (c) Spatial contribution relative to L+PH+R. The horizontal dashed line indicates zero effect for each comparison. Negative values correspond to improved forecasting performance.

Table 5: PatchTST 2×2 configuration study across four datasets (mean $\pm$ std, MSE@720). Sp.% denotes the anchored effect of the spatial module $(+R + S - +R)/\text{Base}$, while +S% denotes the spatial-only effect $(+S - \text{Base})/\text{Base}$.

Dataset	Base	+R	+R+S	+S	Sp.%	+S%
Electricity	0.2238±0.0005	**0.2099±0.0003**	0.2107±0.0015	0.2575±0.0016	+0.4	+15.0
Traffic	**0.5501±0.0019**	0.5918±0.0018	0.5856±0.0010	0.5873±0.0023	−1.1	+6.8
Weather	0.3465±0.0003	0.3424±0.0035	**0.3346±0.0030**	0.3712±0.0195	−2.3	+7.1
ETTh1	0.7226±0.0070	0.7012±0.0042	**0.7009±0.0055**	0.8350±0.0028	−0.0	+15.6

Figure 3 visualizes this control evaluation on the Electricity dataset. The figure shows that the +S configuration consistently degrades forecasting performance across all random seeds, whereas the best performance is obtained when the spatial module is combined with the residual connection (+R+S). These results further highlight the importance of evaluating spatial augmentation together with the accompanying architectural components used in TTS forecasting models.

The Traffic results mirror the NLinear behavior observed earlier in Table 4: the residual alone degrades performance, while the spatial module partially compensates for that degradation. ETTh1 exhibits the strongest +S failure, consistent with its weak cross-channel structure and limited spatial coherence.

Overall, the PatchTST comparisons show that the residual connection plays an important role when applying the spatial module, while the spatial-only configuration consistently reduces forecasting performance in the evaluated settings.

5.3 Stability Across Forecast Horizons

Figure 4 summarizes component effects across forecast horizons for NLinear on Electricity, the representative setting in which both residual and spatial effects remain consistently beneficial at the longest horizon ($H = 720$). Across all hori-

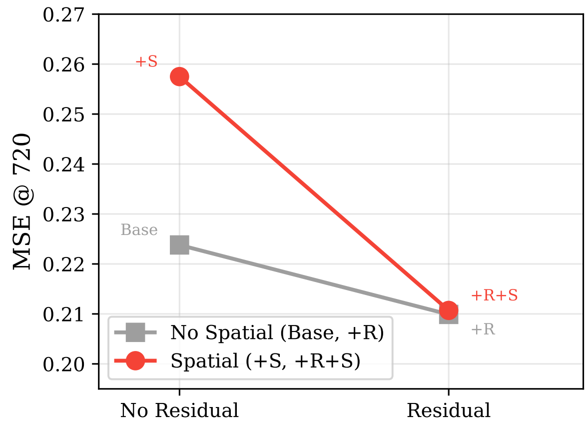

Fig. 3: PatchTST 2×2 interaction on Electricity (MSE@720). The residual path improves forecasting performance both with and without the spatial module, but the largest improvement occurs when the residual and spatial components are combined.

zons, the full TTS architecture improves over the backbone baseline, although the magnitude of the improvement varies with forecasting difficulty.

The residual contribution declines most sharply as the forecast horizon increases, while the spatial contribution remains comparatively stable. At shorter horizons ($H = 96$), the combined residual and spatial correction exceeds 13% relative to L+PH, whereas at $H = 720$ the improvement decreases to approximately 3%. Despite this reduction, both residual and spatial effects remain beneficial once measured relative to L+PH.

Table 6 provides the corresponding numerical results. At shorter horizons, the prediction head alone is mildly harmful, while the residual and spatial components more than offset that degradation. At longer horizons, the prediction head itself becomes beneficial, reducing the relative contribution of the subsequent components.

Table 6: Component effects across forecast horizons for NLinear on Electricity (mean ± std). All percentage effects are normalized by the backbone baseline L.

	MSE				Component Effect (%)			
H	L	L+PH	L+PH+R	L+PH+R+S	PH	Res	Sp.	Gain
96	0.1861±0.0002	0.1919±0.0014	0.1756±0.0003	0.1662±0.0010	+3.1	−8.8	−5.1	−10.7
192	0.1889±0.0003	0.1906±0.0005	0.1840±0.0007	0.1781±0.0008	+0.9	−3.5	−3.1	−5.7
336	0.2074±0.0003	0.2030±0.0015	0.1991±0.0005	0.1935±0.0006	−2.1	−1.9	−2.7	−6.7
720	0.2545±0.0005	0.2366±0.0024	0.2311±0.0009	0.2288±0.0016	−7.0	−2.2	−0.9	−10.1

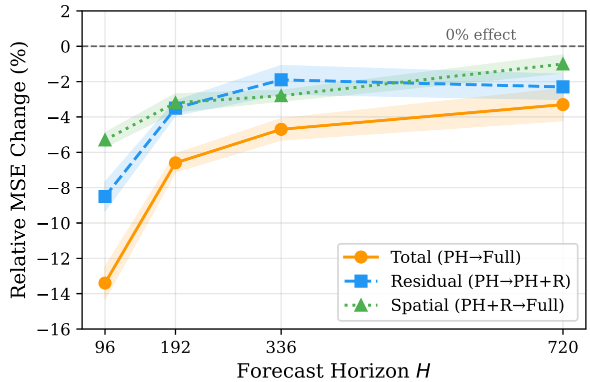

Fig. 4: Component effects across forecast horizons for NLinear on Electricity. The dashed horizontal line marks zero effect for each component-wise comparison. Negative values indicate improved forecasting performance.

Overall, the horizon sweep shows that the relative component behavior remains qualitatively stable across forecasting difficulty: residual and spatial corrections continue to improve performance relative to L+PH, although their magnitude decreases as the forecast horizon increases.

6 Analysis and Discussion

6.1 Backbone and Dataset Dependence

Figure 2 shows that the effect of the spatial module depends strongly on both the dataset and the temporal backbone. These trends are broadly consistent with the dataset profiling statistics in Table 3. Traffic and Electricity, which exhibit stronger cross-channel correlation, generally provide the greatest opportunity for spatial benefit, whereas ETTh1 provides comparatively limited coherent cross-channel structure because of its weak correlations and small number of heterogeneous variables.

However, even when strong cross-channel correlation is present, the direction and magnitude of the effect of the spatial module still depend heavily on the temporal backbone. In particular, the different normalization strategies used by DLinear, NLinear, and RLinear appear to substantially influence how the spatial module behaves. This residual contribution is consistently beneficial on Electricity, mixed on Weather, and highly backbone-dependent on Traffic. Spatial effects exhibit an even stronger dependence on the backbone: NLinear generally benefits from the spatial module, DLinear shows mixed behavior, and RLinear is consistently degraded by the spatial module.

These comparisons show that the effect of spatial augmentation cannot be interpreted independently of the surrounding architectural components. In several settings, the residual connection accounts for most of the improvement associated with the full TTS configuration, while the spatial module provides only modest additional benefit. In other settings, the spatial module partially compensates for degradation introduced by the residual path. The prediction head alone also frequently improves over the backbone baseline before either residual fusion or spatial augmentation is introduced.

Overall, the results (Table 4) suggest that the effectiveness of spatial augmentation in LTSF depends jointly on dataset structure, backbone normalization, and the accompanying architectural components used within the TTS configuration. More broadly, the findings indicate that successful spatial learning in LTSF depends not only on the availability of cross-channel structure, but also on how effectively the temporal backbone exposes that structure to the spatial module.

6.2 Spatial Module as Residual Correction

Table 5 and Figure 3 show that the effect of the spatial module in PatchTST depends strongly on the residual connection. Applying the spatial module without residual fusion (+S) consistently degrades forecasting performance across all four datasets, with the largest degradation occurring on Electricity and ETTh1.

By contrast, the combined residual and spatial configuration (+R+S) remains competitive and improves performance on Traffic and Weather while remaining approximately neutral on Electricity and ETTh1. These comparisons indicate that the spatial module behaves differently when applied together with residual fusion than when applied in isolation.

More broadly, the results suggest that residual connections play an important role when applying spatial augmentation within TTS forecasting architectures. The findings also reinforce the importance of evaluating spatial modules together with the accompanying architectural components rather than only through backbone versus the full model comparisons.

6.3 Forecast Horizon and Context Effects

Figure 4 shows that both residual and spatial effects weaken as the forecast horizon increases. The full TTS configuration remains beneficial across all horizons for NLinear on Electricity, but the magnitude of the improvement decreases substantially from short- to long-term forecasting.

The residual effect declines most sharply with horizon length, whereas the spatial effect remains comparatively stable. At shorter horizons, the prediction head alone is mildly harmful, and the residual and spatial components offset that degradation. At longer horizons, the prediction head itself becomes beneficial, reducing the relative effect of the subsequent components.

The PatchTST comparisons at two input lengths show a similar trend. Increasing the input window from $L = 96$ to $L = 336$ improves the combined residual and spatial configuration, while the spatial-only configuration remains harmful at both lengths. These observations suggest that the effectiveness of spatial augmentation depends on both forecasting horizon and input context.

6.4 Architectural Implications

Several implications for the design of spatiotemporal models emerge from this analysis. Historically, STGNNs have relied on complex temporal components such as TCNs or RNNs. Our findings show that simple linear backbones can also serve as effective temporal foundations for lightweight TTS architectures in long-term forecasting. This suggests that the role of the temporal backbone in TTS architectures extends beyond temporal forecasting accuracy alone, influencing the representations available for subsequent spatial processing. This is particularly relevant in modern LTSF settings, where efficient architectures are often preferred because large spatio-temporal models can introduce substantial computational cost and optimization complexity.

At the same time, the effectiveness of spatial augmentation depends strongly on both the backbone and the dataset. Traffic and Electricity, which exhibit stronger cross-channel correlation, show the largest potential for spatial benefit, but the direction and magnitude of the effect still depend heavily on the temporal backbone, including the normalization strategy used by each model. As noted in Section 5.1, the strongest overall forecasting performance across all datasets was achieved by RLinear configurations without the full spatial configuration. Conclusions drawn from a single backbone therefore do not reliably generalize across architectures and datasets.

The results also highlight the importance of evaluating spatial augmentation together with the accompanying architectural components, particularly residual connections and prediction heads. Across several settings, much of the improvement associated with the full TTS configuration is already present before the spatial module is introduced. More broadly, the experiments suggest that lightweight spatial augmentation is most effective when applied alongside residual fusion rather than in isolation.

6.5 Limitations and Future Directions

Our findings characterize one specific type of spatial augmentation: lightweight learned adjacency spatial modules applied within a TTS architecture. Alternative designs, such as attention-based channel mixing, alternative graph neural network (GNN) architectures, or more highly parameterized spatial components, may behave differently. Furthermore, datasets that provide predefined spatial graphs, such as physical sensor networks, might exhibit substantially stronger spatial effects than the standard benchmarks evaluated here.

The experiments focus primarily on MSE at $H = 720$, with horizon sweeps over $H \in 96, 192, 336, 720$ and limited input length variation. Additionally, the isolated spatial configuration (+S) was evaluated only with PatchTST; extending this configuration analysis to the linear backbones would further broaden the architectural comparison across model classes. Similarly, analyzing whether the adjacency structures capture semantically meaningful relationships could help connect predictive utility to interpretability.

Finally, the systematic architectural evaluation used in this work can be extended to other channel-dependent forecasting models. Attention based architectures such as iTransformer [16] and cross-channel mixers such as TSMixer [5] could be analyzed using analogous comparisons that isolate cross-channel processing from the remaining forecasting pipeline. Broader architectural comparisons across diverse spatial designs remain a valuable direction for future LTSF research.

7 Conclusion

This work presented a systematic evaluation of spatial modules for long-term multivariate time series forecasting (LTSF). We analyzed the effect of adding lightweight spatial modules to DLinear, NLinear, RLinear, and PatchTST across four benchmark datasets using multiple architectural configurations within a shared TTS design.

The experiments show that the effectiveness of additional spatial module depends strongly on the temporal backbone, dataset, and architectural configuration. While spatial modules can improve forecasting performance in several settings, the strongest overall configurations are not consistently associated with the full spatial architecture. In many cases, prediction heads and residual connections account for a substantial portion of the observed improvement, while the effect of the spatial module itself varies considerably across backbones and datasets.

Overall, the results highlight the importance of evaluating spatial augmentation together with the accompanying architectural components used to integrate it. As lightweight temporal backbones continue to advance in LTSF, understanding how temporal representations interact with spatial augmentation behaves across different architectures and datasets may be as important as the design of the spatial modules themselves.

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